

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Monday, 24 August 2026, 9:02 PM**State** Finished**Completed on** Monday, 24 August 2026, 9:03 PM**Time taken** 1 min 26 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: 3

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: 2

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int find_majority_element(int *nums, int n) {
5      int candidate = nums[0];
6      int count = 1;
7
8      for (int i = 1; i < n; i++) {
9          if (count == 0) {
10             candidate = nums[i];
11             count = 1;
12         } else if (nums[i] == candidate) {
13             count++;
14         } else {
15             count--;
16         }
17     }
18     return candidate;
19 }
20
21
22 int main() {
23     int n;
24     if (scanf("%d", &n) != 1) return 0;
25
26     int *nums = (int *)malloc(n * sizeof(int));
27     if (nums == NULL) return 0;
28
29     for (int i = 0; i < n; i++) {
30         if (scanf("%d", &nums[i]) != 1) {
31             free(nums);
32             return 0;
33         }
34     }

```

```
35
36     int result = find_majority_element(nums, n);
37     printf("%d\n", result);
38
39     free(nums);
40     return 0;
41 }
42
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)